Credit Card Fraud Detection

Model Comparison using Logistic Regression, Decision Tree, and MLP (ANN)

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Abstract

This mini-project builds and compares three models—Logistic Regression, Decision Tree, and a Multi-Layer Perceptron (ANN)—for credit card fraud detection. Using a Kaggle dataset, we implement preprocessing, train–test split, SMOTE for handling imbalance, training, and evaluation using Precision, Recall, F1-score, and ROC-AUC. Results show that the ANN attains the best overall performance.

1 Problem Statement

Credit card fraud is a major challenge for financial institutions. The goal is to classify transactions as *fraudulent* or *legitimate* using machine learning. We evaluate three models aligned with the syllabus:

- Logistic Regression (baseline linear classifier),
- Decision Tree (interpretable, non-linear),
- Multi-Layer Perceptron (ANN).

2 Dataset

We use the "Credit Card Fraud Detection Dataset 2023" from Kaggle [1]. Features: anonymized variables $V1$–$V28$, transaction `Amount`, and target label `Class` (fraud=1, non-fraud=0). The `id` column was dropped since it is not predictive.

3 Methodology

The pipeline (Figure 1) includes:

1. Preprocessing: scaling numerical features.
2. Train/test split (80/20 stratified).
3. SMOTE oversampling on training data to handle class imbalance.
4. Model training: Logistic Regression, Decision Tree, ANN.
5. Evaluation: Precision, Recall, F1-score, ROC-AUC.

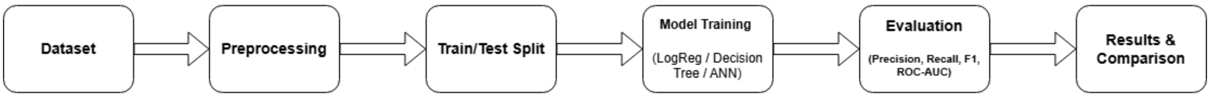

Figure 1: Workflow for Credit Card Fraud Detection.

4 Code Snippet

```
logreg = ImbPipeline([
  ('pre', preprocess),
  ('smote', SMOTE(random_state=42)),
  ('clf', LogisticRegression(class_weight='balanced',
                             solver='liblinear',
                             max_iter=1000))
])

tree = ImbPipeline([
  ('pre', preprocess),
  ('smote', SMOTE(random_state=42)),
  ('clf', DecisionTreeClassifier(class_weight='balanced',
                                 random_state=42))
])

mlp = ImbPipeline([
  ('pre', preprocess),
  ('smote', SMOTE(random_state=42)),
  ('clf', MLPClassifier(hidden_layer_sizes=(64,32),
                        activation='relu',
                        early_stopping=True,
                        max_iter=120))
])
```

5 Results

Table 1 shows the evaluation metrics of all three models on the test set. The ANN clearly achieves the highest values in Precision, Recall, F1-score, and ROC-AUC, indicating its superior performance. Logistic Regression performs slightly lower, while Decision Tree is very strong and close to ANN.

Table 1: Model comparison on test set.

Model	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.9773	0.9522	0.9646	0.9935
Decision Tree	0.9968	0.9987	0.9978	0.9978
MLP (ANN)	**0.9991**	**1.0000**	**0.9996**	**0.99996**

From Table 1, we see that the ANN achieves almost perfect scores with Recall = 1.0, meaning no fraudulent transaction was missed. Decision Tree also performs very well but is slightly less accurate than ANN. Logistic Regression, being linear, is less powerful but still a good baseline.

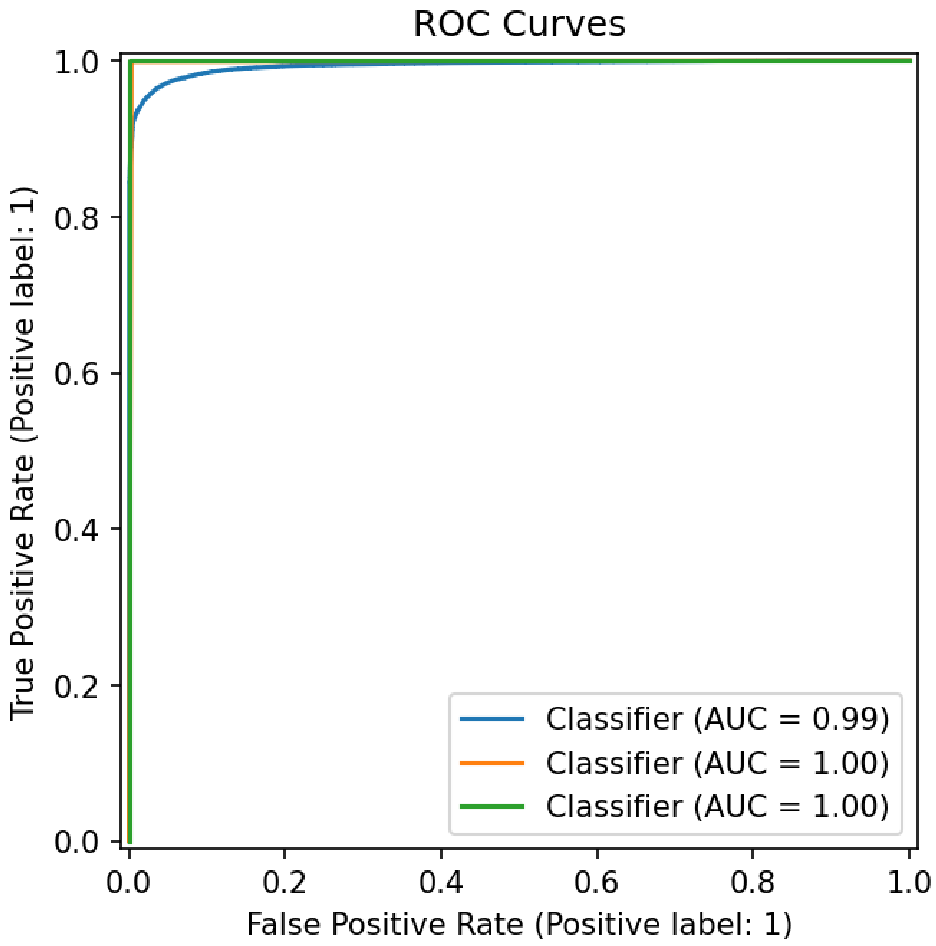

Figure 2: ROC curves for the three models.

Figure 2 shows the ROC curves. The curve for ANN is almost touching the top-left corner, which indicates near-perfect classification ability. Decision Tree is also close, while Logistic Regression is slightly below but still strong. A higher ROC curve means better ability to distinguish fraud vs non-fraud.

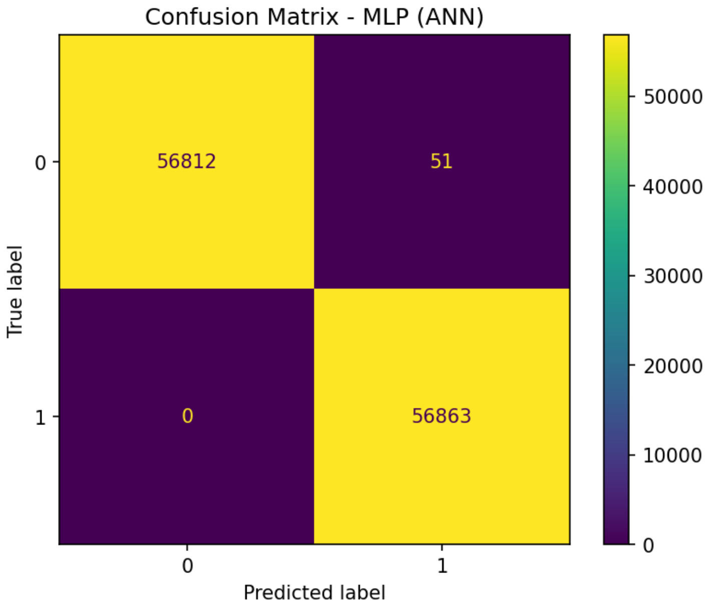

Figure 3: Confusion matrix for best model (ANN).

Figure 3 presents the confusion matrix for ANN. It shows that almost all legitimate and fraudulent transactions were classified correctly. Only a very small number of false positives occurred (legitimate marked as fraud), while false negatives (fraud missed) are zero. This is desirable in fraud detection since missing frauds is more dangerous than raising a few false alarms.

6 Future Scope

This project can be extended in several directions:

- Deploy the trained model as a REST API for real-time fraud detection in banking apps.
- Experiment with advanced models like Random Forests, Gradient Boosted Trees, or Transformer-based models.
- Perform cost-sensitive learning, where missing a fraud is penalized more than a false alarm.
- Explore feature importance to gain insights into which transaction attributes contribute most to fraud detection.

7 Conclusion

The MLP (ANN) outperformed Logistic Regression and Decision Tree, achieving nearly perfect recall and ROC-AUC. Decision Tree offers interpretability, while Logistic Regression provides a strong linear baseline. The project demonstrates that deep learning yields the highest fraud detection performance for this dataset.

References

[1] Gayan Nuwani Nelgiriyewithana, “Credit Card Fraud Detection Dataset 2023,” Kaggle, 2023. Available: https://www.kaggle.com/datasets/nelgiriyewithana/credit-card-fraud-detection-dataset-2023

[2] Pedregosa, F. et al., “Scikit-learn: Machine Learning in Python,” JMLR, vol. 12, pp. 2825–2830, 2011.

[3] Lemaître, G., Nogueira, F., and Aridas, C.K., “Imbalanced-learn: A Python Toolbox to Tackle the Curse of Imbalanced Datasets,” JMLR, vol. 18, no. 17, pp. 1–5, 2017.